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In the same way businesses use big data to pursue profits, governments use it to promote the public good.

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Big-Data Applications in the Government Sector

BIG DATA, a general term for the massive amount of digital data being collected from all sorts of sources, is too large, raw, or unstructured for analysis through conventional relational database techniques. Almost 90% of the world's data today was generated during the past two years, with 2.5 quintillion bytes of data added each day.⁷ Moreover, approximately 90% of it is unstructured. Still, the overwhelming amount of big data from the Web and the cloud offers new opportunities for discovery, value creation, and rich business intelligence for decision support in any organization. Big data also means new challenges involving complexity, security, and risks to privacy, as well as a need for new technology and human skills.

Big data is redefining the landscape of data management, from extract, transform, and load, or ETL, processes to new technologies (such as Hadoop) for cleansing and organizing unstructured data in big-data applications.

Although the business sector is leading big-data-application development, the public sector has begun to derive insight to help support decision making in real time from fast-growing in-motion data from multiple sources, including the Web, biological and industrial sensors, video, email, and social communications.⁸ Many white papers, journal articles, and business reports have proposed ways governments can use big data to help them serve their citizens and overcome national challenges (such as rising health care costs, job creation, natural disasters, and terrorism).⁹ There is also some skepticism as to whether it can actually improve government operations, as governments must develop new capabilities and adopt new technologies (such as Hadoop and NoSQL) to transform it into information through data organization and analytics.⁴

Here, we ask whether governments are able to implement some of today's big-data applications associated with the business sector. We first compare the two sectors in terms of goals, missions, decision-making processes, decision actors, organizational structure, and strategies (see the table here), then turn to several current applications in technologically advanced

» key insights

- Businesses, governments, and the research community can all derive value from the massive amounts of digital data they collect.
- Governments of leading ICT countries have initiated big-data application projects to enhance operational efficiency, transparency, citizens' well-being and engagement in public affairs, economic growth, and national security.
- Analyzing big-data application projects by governments offers guidance for follower countries for their own future big-data initiatives.



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